Forecasting Conflict Fatalities with a Democracy-Based Model

Conflict Forecasting Project
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Abstract

This paper evaluates whether a democracy-centered specification improves forecasts of logged conflict fatalities over forecast horizons of one to seven months. Drawing on ten indicators from V-Dem, the Worldwide Governance Indicators, Freedom House, and Polity, the analysis estimates two machine-learning models: Random Forest and Gradient Boosting Machine (GBM). The results are clear. Democracy-based models do not outperform the benchmark specifications, especially the lag-based benchmark, which remains the strongest model across MSE, RMSE, and both deterministic and probabilistic CRPS. Among the democracy models, however, GBM consistently performs better than Random Forest. Substantively, the findings suggest that while democratic institutions capture meaningful aspects of political order, they add less predictive leverage than recent conflict dynamics.

1. Introduction

Can democratic institutions improve forecasts of future conflict fatalities? This paper addresses that question by developing and testing a democracy-based forecasting specification against benchmark models built earlier in the course.

The underlying intuition is straightforward: regimes characterized by broader participation, stronger accountability, more effective institutions, and greater legal constraint may be better able to channel conflict through nonviolent mechanisms. If so, indicators of democracy should contain predictive information about future conflict intensity.

To evaluate this claim, I estimate two models built on democracy-related predictors and compare their out-of-sample performance to benchmark forecasting models. The goal is not only to assess whether democratic institutions matter substantively, but whether they improve predictive accuracy in practice.

2. Theory and Operationalization

The democracy specification is rooted in the argument that political inclusion, institutional constraint, and civil liberty can reduce the likelihood that conflict escalates into higher levels of violence. Democracies may create more opportunities for contestation within formal institutions, lower incentives for violent escalation, and strengthen state capacity to manage social and political tensions.

To capture these dimensions, I use ten indicators drawn from major cross-national data sources. Together, they represent electoral competition, liberal protections, participation, deliberation, equality, accountability, government effectiveness, rule of law, civil liberties, and regime type.

Table 1: Democracy specification

Variable	Source	Concept captured
`vdem_v2x_polyarchy`	V-Dem	Electoral contestation and procedural democracy
`vdem_v2x_libdem`	V-Dem	Liberal protections and institutional checks on power
`vdem_v2x_partipdem`	V-Dem	Participatory inclusion and political voice
`vdem_v2x_delibdem`	V-Dem	Deliberative quality of decision-making
`vdem_v2x_egaldem`	V-Dem	Equality of political influence and access
`wgi_VA_EST`	WGI	Voice and accountability
`wgi_GE_EST`	WGI	Government effectiveness
`wgi_RL_EST`	WGI	Rule of law and legal constraint
`fh_cl`	Freedom House	Civil liberties and associational freedom
`polity_polity2`	Polity	Broad regime-type orientation

This specification is intentionally theory-driven. Rather than selecting variables simply because they are available, the model groups indicators that plausibly capture the institutional channels through which democracy may shape conflict trajectories.

3. Research Design

The outcome is logged conflict fatalities at seven future horizons, measured by the variables `ln_ged_best_sb_s1` through `ln_ged_best_sb_s7`. The analysis follows the standard course train/test split: the training data end at month 372, while the test period covers months 373 through 420.

I estimate two democracy-based forecasting models:

1. **Democracy Random Forest**, which is well suited to nonlinearities and interaction effects with minimal parametric structure.

2. **Democracy GBM**, which incrementally improves fit by combining many weak learners and often performs well when predictive signal is diffuse.

These models are evaluated against two benchmark specifications:

- a lag-based benchmark random forest,
- a rolling-average benchmark random forest.

Performance is assessed using matched prediction rows and standard out-of-sample metrics: mean squared error (MSE), root mean squared error (RMSE), deterministic CRPS, and probabilistic CRPS. I also examine 80% interval coverage and interval sharpness to assess the quality of the uncertainty estimates.

4. Results

The central finding is unambiguous: democracy-based models do not outperform the benchmark models. The lag-based benchmark remains the strongest specification across forecast horizons and evaluation metrics.

At the same time, the democracy models are not equivalent. Across nearly all horizons, the Democracy GBM performs better than the Democracy Random Forest, suggesting that boosted trees are better able to extract whatever predictive signal exists in the democracy indicators. Even so, that advantage is not large enough to surpass the predictive power of recent conflict dynamics embedded in the benchmark models.

Figure 1 presents the full comparison across horizons. The pattern is consistent: benchmark performance remains strongest, Democracy GBM ranks ahead of Democracy Random Forest, and the democracy specifications fall short of the leading benchmark model.

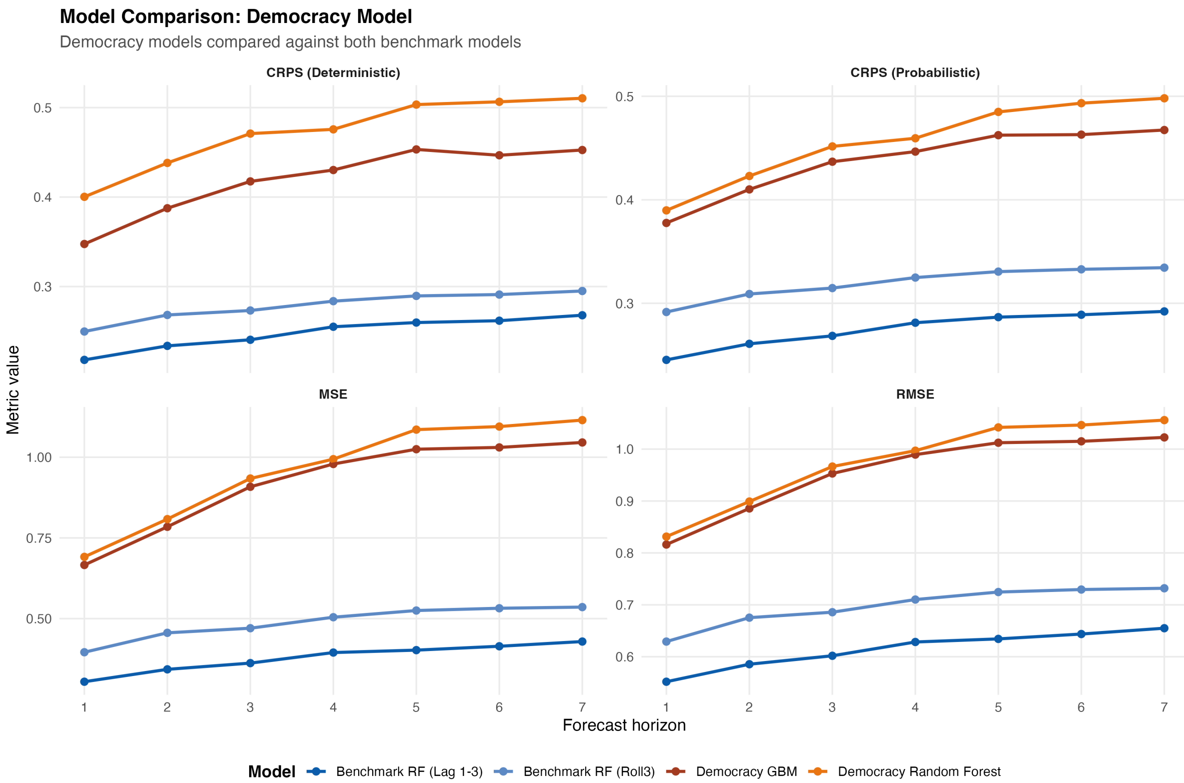

Figure 1: Model performance across forecast horizons

Table 2 reinforces this conclusion by reporting the best observed score at each horizon for each metric. In every case, the winning model is the lag-based benchmark random forest.

Table 2: Best score by horizon and metric

Horizon	MSE	RMSE	CRPS (Det.)	CRPS (Prob.)
1	0.305	0.552	0.218	0.245
2	0.343	0.586	0.234	0.261
3	0.362	0.602	0.240	0.269
4	0.395	0.628	0.255	0.281
5	0.402	0.634	0.260	0.287
6	0.414	0.644	0.262	0.289
7	0.429	0.655	0.268	0.292

Note: At every forecast horizon, the best-performing model is the Benchmark RF with Lag 1–3 predictors.

5. Variable Importance

Although the democracy models do not lead in predictive performance, the feature-importance results remain substantively informative. The most influential predictors are consistently tied to accountability, legal order, and regime structure. In particular, voice and accountability, rule of law, and the Polity score appear repeatedly near the top of the rankings.

This pattern suggests that democracy-related variables do capture relevant features of political environments associated with violence risk. Their limitation is not conceptual irrelevance, but weaker predictive leverage relative to recent conflict history.

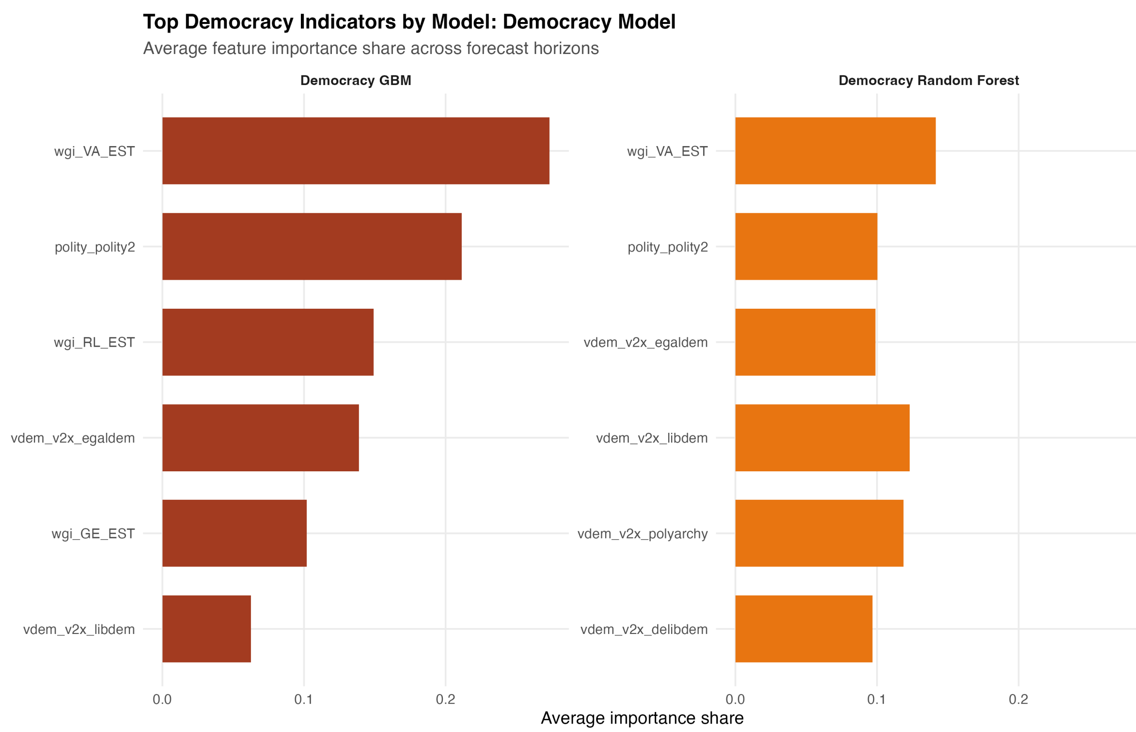

Figure 2: Top democracy predictors by average importance share

6. Conclusion

This paper tested whether a democracy-centered specification improves forecasts of conflict fatalities. The answer is no: democracy-based models do not outperform the benchmark models, and the lag-based benchmark remains superior across all principal predictive metrics.

Still, the exercise yields two useful conclusions. First, among the democracy specifications, GBM is consistently stronger than Random Forest, indicating that model choice matters even within the same theoretical family. Second, democracy indicators retain substantive value, especially those related to accountability and rule of law, but their predictive contribution is weaker than the signal contained in recent patterns of violence.

More broadly, the findings underscore a recurring lesson in conflict forecasting: theoretically meaningful variables do not always translate into the strongest predictive models. In this case, recent conflict dynamics remain more informative about near-term fatalities than broader institutional characteristics of democracy.